

The Locally Nilpotent Part of Nonboundary Localization

Quotients

G_c is never simple: a weight filtration with simple quotients,
zero socle, and infinite length

Progress report

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1 Definitions, notation, and assumptions

Affine algebra and induced modules. All vector spaces and tensor products are over $\mathbb{C}$. We use the affine Lie algebra without the degree derivation:

$$\widehat{\mathfrak{sl}}_2 = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K, \quad U = U(\widehat{\mathfrak{sl}}_2).$$

Write $x_i = x \otimes t^i$. For $i, j \in \mathbb{Z}$, the brackets are

$$\begin{aligned} [e_i, f_j] &= h_{i+j} + i\delta_{i+j,0}K, & [h_i, h_j] &= 2i\delta_{i+j,0}K, \\ [h_i, e_j] &= 2e_{i+j}, & [h_i, f_j] &= -2f_{i+j}, \end{aligned} \tag{1}$$

with K central and $[e_i, e_j] = [f_i, f_j] = 0$. Fix $a, b \in \mathbb{Z}$ with $a + b \geq 0$ and put $L = a + b + 1 \geq 1$. Set

$$S_{a,b} = \text{span}_{\mathbb{C}}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}.$$

This is a Lie subalgebra. A character $\chi : S_{a,b} \rightarrow \mathbb{C}$ is specified by $c, \lambda_0, \dots, \lambda_L \in \mathbb{C}$ through

$$\chi(K) = c, \quad \chi(h_i) = \lambda_i \ (0 \leq i \leq L), \quad \chi(h_i) = 0 \ (i > L), \quad \chi(e_i) = \chi(f_j) = 0,$$

with $\lambda_i := 0$ for $i > L$. Define

$$\mathbb{C}_\chi = \mathbb{C}1_\chi, \quad M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_\chi, \quad u = 1 \otimes 1_\chi.$$

By [?, Theorem 1.1],

$$M_{a,b}(\chi) \text{ is simple} \iff \lambda_L \neq 0 \text{ and } c \neq -2.$$

This criterion is not used in Sections 2–4 below.

Deep-region localization. Fix $c < b$ and set $F = f_c$. The derivation $\text{ad } F$ is locally nilpotent on U , so the Ore localization $U_F = U[F^{-1}]$ exists and

$$\mathcal{D}_F M = U_F \otimes_U M, \quad T_F M = \mathcal{D}_F M / M.$$

PBW with F first makes M free over $\mathbb{C}[F]$, so F acts injectively and the quotient notation is legitimate. The deep region is

$$c \leq -a - 1, \quad \text{equivalently} \quad d_* := b - c \geq L.$$

The spectral-flow automorphism σ_{-c} , defined on generators by $e_i \mapsto e_{i-c}$, $f_i \mapsto f_{i+c}$, $h_i \mapsto h_i - c\delta_{i,0}K$, identifies $T_F M_{a,b}(\chi)$ with

$$Q = \mathcal{T}_{f_0} M_{A,B}(\chi'), \quad A = a + c \leq -1, \quad B = b - c = d_* \geq L, \quad \mu := \chi'(h_0) = \lambda_0 - c\kappa.$$

Write

$$E = e_0, \quad H = h_0, \quad w_m = f_0^{-m}u + M \quad (m \geq 1).$$

Then $[E, f_0] = H$, $Eu = 0$, $Hu = \mu u$, so (E, H, f_0) is an $\mathfrak{sl}_2$ -triple with u of weight μ .

PBW notation. Let $\mathcal{X}$ be the set of ordered monomials

$$\mathcal{X} = \left\{ X = f_{j_1} \cdots f_{j_p} e_{i_1} \cdots e_{i_q} h_{r_1} \cdots h_{r_s} : j_t \leq B, j_t \neq 0, i_t \leq A, r_u < 0 \right\},$$

including the empty word, with $p, q, s \geq 0$ arbitrary and the factors in the displayed order. Put

$$\deg X = 2q - 2p,$$

so that $H(F^{-m}Xu) = (\mu + 2m + \deg X)F^{-m}Xu$. Then

$$\{F^{-m}Xu : m \geq 1, X \in \mathcal{X}\}$$

is a basis of Q , and the weight spaces are Q_ν , $\nu \in \mu + 2\mathbb{Z}$.

The locally nilpotent part. Set

$$G = G_c := \{v \in Q : E^n v = 0 \text{ for some } n \geq 1\}, \quad P_\nu := \ker E \cap G \cap Q_\nu.$$

Because $\text{ad } E$ is locally nilpotent, G is a $\widehat{\mathfrak{sl}}_2$ -submodule of Q , and every nonzero $\widehat{\mathfrak{sl}}_2$ -submodule of G contains a nonzero vector of some P_ν : apply powers of E to a nonzero vector and take its last nonzero image. In the shallow region $-a \leq c < b$ the mode $E = e_{-c}$ is injective on Q , so $G = 0$; the theory below is the complete answer for all nonboundary indices.

2 The E-action and the primitive space

Lemma 1. *On the PBW basis of Q ,*

$$E(F^{-m}Xu) = F^{-m}T(X)u - m(\mu + \deg X + m + 1)F^{-m-1}Xu, \quad T := \pi_0 \circ \text{ad}(e_0), \quad (2)$$

where π_0 projects M onto its F^0 -part. The operator T sends $\mathcal{X}$ -words to combinations of $\mathcal{X}$ -words, raises $\deg$ by 2, and preserves the annihilating ideal: $Wu = 0 \Rightarrow T(W)u = 0$.

Proof. The triple identity $EF^{-m} = F^{-m}E - mF^{-m-1}H - m(m+1)F^{-m-2}F$ and $Eu = 0$ give $E(F^{-m}Xu) = F^{-m}(\text{ad } E)(X)u - m(\mu + \deg X)F^{-m-1}Xu - m(m+1)F^{-m-2}FXu$. In the localization $F^{-m-2}FXu = F^{-m-1}Xu$, and $(\text{ad } E)(X)u \in M$ contributes to Q only through its F^0 -part. The word-level description follows from $[e_0, f_j] = h_j$, $[e_0, h_r] = -2e_r$, $[e_0, e_i] = 0$, and the last assertion from $T(W)u = \pi_0 E(Wu)$. $\square$

Lemma 2. *If $0 \neq v \in Q_\nu$ and $Ev = 0$, then $\nu \in \mathbb{Z}_{\geq 0}$ and the highest pole of v in the PBW basis is exactly $\nu + 1$.*

Proof. Write $v = \sum_{m=1}^N F^{-m}X_m u$ with $\deg X_m = \nu - \mu - 2m$. By (??),

$$Ev = \sum_m F^{-m}T(X_m)u - \sum_m m(\nu - m + 1)F^{-m-1}X_m u.$$

At pole $N + 1$ only the diagonal survives, so $Ev = 0$ forces $N = \nu + 1$ and $\nu \geq 0$, with $X_{\nu+1}$ unconstrained. $\square$

Theorem 3. *For $\nu \in \mathbb{Z}_{\geq 0}$,*

$$P_\nu \cong \left\{ X \in \mathcal{X} : \deg X = -(\mu + \nu + 2), \quad T^{\nu+1}(X)u = 0 \right\}, \quad (3)$$

the isomorphism being $v \mapsto$ its top-pole layer at pole $\nu + 1$. In particular $P_\nu \neq 0$ forces $\mu + \nu$ even, and every nonzero P_ν is infinite-dimensional: multiply top layers by powers of h_{-1} .

Proof. Equating the coefficient of $F^{-m}X$ in $Ev = 0$ gives

$$T(X_m)u = (m-1)(\nu-m+2)X_{m-1}u, \quad X_0 = 0, \quad X_{\nu+2} = 0. \quad (4)$$

For $m \geq 2$ the scalar $(m-1)(\nu-m+2) \neq 0$, so the layers $X_\nu, \dots, X_1$ are determined one by one from the top layer: a strictly decreasing pole with a unique solution. The remaining condition at $m = 1$ is $T(X_1)u = 0$, which by the recursion is exactly $T^{\nu+1}(X_{\nu+1})u = 0$. $\square$

Corollary 4. (i) If $\mu = -2$, then $P_0 = \mathbb{C}w_1$: the top layer is the empty word, and for $s \geq 1$ one has $T(h_{-1}^s)u = -2sh_{-1}^{s-1}e_{-1}u \neq 0$.

(ii) In the basic module $M_{-1,1}$ at $\lambda_0 = 0$ the vector $e_0^2 f_1^2 w_1$ lies in P_2 with top layer f_1^2 at pole 3, since $T^3(f_1^2)u = (-12h_1 e_1 + 4e_2)u = 0$.

3 The $\mathfrak{sl}_2$ -decomposition and the projections

Theorem 5. Every vector of G is contained in a finite-dimensional $\mathfrak{sl}_2(E, H, f_0)$ -submodule, and

$$G \cong \bigoplus_{\nu \in \mathbb{Z}_{\geq 0}} P_\nu \otimes L(\nu), \quad (5)$$

where $L(\nu)$ is the $(\nu+1)$ -dimensional irreducible $\mathfrak{sl}_2$ -module. Thus the E -primitive vectors of G have weights in $\mathbb{Z}_{\geq 0}$, and P_ν is the multiplicity space of $L(\nu)$.

Proof. E is locally nilpotent on G by definition, and f_0 is locally nilpotent on all of Q . Hence the $\mathfrak{sl}_2$ -span of any vector is spanned by the finitely many vectors $f_0^s E^s v$, and complete reducibility is Weyl's theorem. A primitive vector p with $f_0^N p = 0$ generates $L(\nu)$, ν its H -weight, and $f_0^N p = 0$ forces $\nu \leq N-1$, so $\nu \in \mathbb{Z}_{\geq 0}$. $\square$

Let $p \in P_\nu$ have top layer X , and write X_m for its layers.

Proposition 6. (i) For $i \leq A$: $e_i p \in P_{\nu+2}$, and its top layer is $e_i X$. Write $\hat{e}_i(X) = e_i X$.

(ii) For $r < 0$ and $\nu \geq 1$: $h_r p + \frac{2}{\nu+2} f_0 e_r p \in P_\nu$, with top layer

$$\hat{h}_r(X) = \frac{\nu}{\nu+2} \left(-h_r X + \frac{1}{\nu} f_r T(X) \right).$$

For $\nu = 0$ one has $h_r p = -f_0 e_r p$, a vector of the $L(2)$ -block of $e_r p \in P_2$, so the P_0 -part of $h_r p$ is zero.

(iii) For $j \geq 1$: $E^2 f_j p = -2e_j p$, and the primitive part of $f_j p$ in $P_{\nu+2}$ has top layer $e_j X$ (lower layers by the recursion).

Moreover G is generated, as a $\widehat{\mathfrak{sl}_2}$ -module, by $P = \bigoplus_\nu P_\nu$, and the submodule generated by any $0 \neq p \in P_\nu$ contains $L(\nu)p$ together with the images of p under all words in $\{\hat{e}_i, \hat{h}_r, \hat{f}_j, E, f_0\}$.

Proof. (i) $[E, e_i] = 0$, and the pole analysis of Lemma ?? applied to $e_i p$ gives the top layer $e_i X$: the corrections of $[e_i, f_0^{-m}]$ land at poles $m+1, m+2$ and feed the lower layers. (ii) $[E, h_r] = -2e_r$ and $[f_0, e_r] = -h_r$ for $r < 0$ give $h_r p = e_r f_0 p - f_0 e_r p$. For $\nu = 0$: $f_0 p = 0$, so $h_r p = -f_0 e_r p$. For $\nu \geq 1$: $E f_0 e_r p = (\nu+2)e_r p$, hence $E(h_r p + \frac{2}{\nu+2} f_0 e_r p) = 0$. The pole- $(\nu+2)$ layers cancel exactly, because $[e_r, f_0^{-m}] = -m f_0^{-m-1} h_r - m(m+1) f_0^{-m-2} f_r$. (iii) $[E, f_j] = h_j$ gives $E^2 f_j p = -2e_j p$; then $[E, e_j] = 0$ gives $e_j p \in P_{\nu+2}$ (here $j > A$, since $A \leq -1$). $\square$

4 The weight filtration

For $\nu \geq 0$ define

$$G^{\geq \nu} := \bigoplus_{\nu' \geq \nu} P_{\nu'} \otimes L(\nu'),$$

the span of the $\mathfrak{sl}_2$ -blocks of top weight at least ν .

Lemma 7. *Each $G^{\geq \nu}$ is a $\widehat{\mathfrak{sl}}_2$ -submodule of G . Moreover $P_{\nu} \neq 0$ if and only if $\nu \geq \nu_0$ and $\nu \equiv \mu \pmod{2}$, where ν_0 is the smallest $\nu \geq 0$ admitting a top layer, and for every such ν the map $e_1 : P_{\nu} \rightarrow P_{\nu+2}$ is nonzero.*

Proof. The weight never decreases: e_r sends $P_{\nu'}$ to $P_{\nu'+2}$; h_r and f_j have a $P_{\nu'}$ -part (the projections of Proposition ??) and a $P_{\nu'+2}$ -part; E, H, f_0 act inside each block and K is a scalar. Hence $G^{\geq \nu}$ is stable. The parity of ν is forced by $\deg X = -(\mu + \nu + 2)$. For $0 \neq p \in P_{\nu}$, the vector $e_1 p$ has top layer $f_1 X \neq 0$, so the support has no gaps above ν_0 . $\square$

Theorem 8. *With ν_0 as in Lemma ??,*

$$G = G^{\geq \nu_0} \supsetneq G^{\geq \nu_0+1} \supsetneq G^{\geq \nu_0+2} \supsetneq \dots$$

is an infinite strictly descending chain of $\widehat{\mathfrak{sl}}_2$ -submodules (steps with $P_{\nu} = 0$ collapse), and:

(i) *for $\nu \geq 1$,*

$$G^{\geq \nu} / G^{\geq \nu+1} \cong \bigoplus_{q \geq q_{\nu}} (P_{\nu}^{(q)} \otimes L(\nu)), \quad P_{\nu}^{(q)} = \left\{ X \in \mathcal{X} : p(X) = q + \frac{\mu + \nu + 2}{2}, e(X) = q, T^{\nu+1}(X)u = 0 \right\},$$

and each summand is a simple $\widehat{\mathfrak{sl}}_2$ -module;

(ii) *if $P_0 \neq 0$, then $G^{\geq 0} / G^{\geq 1} \cong P_0 \otimes L(0)$ is a direct sum of one-dimensional trivial $\widehat{\mathfrak{sl}}_2$ -modules.*

Consequently G is not simple, $\text{soc}(G) = 0$ (and a fortiori $\text{soc}(Q_c) = 0$), G has no finite composition series, and the head of G is the ν_0 -quotient.

Proof. Properness and nonemptiness of each step follow from Lemma ??. In the quotient $G^{\geq \nu} / G^{\geq \nu+1}$ the mode e_r sends the top vectors of the summands into $G^{\geq \nu+2}$ and acts on lower block levels through the projections $\hat{h}_r, \hat{f}_r$, via $[e_r, f_0^s] = sh_r f_0^{s-1} - s(s-1)f_r f_0^{s-2}$; so each summand is preserved. The projections $\hat{h}_r, \hat{h}_j, \hat{f}_j$ preserve the counts (p, e) of the top layer: their candidate lower-count terms cancel inside P_{ν} by the degree condition $\deg X = -(\mu + \nu + 2)$.

For triangular generation in the h-degree: on the summand of type $P_{\nu}^{(q)}$, the projection $\hat{h}_r$ acts as $-\frac{\nu}{\nu+2} h_r X$ plus terms of lower h-degree or higher e-degree, and the leading scalar is nonzero for $\nu \geq 1$. Hence from any nonzero class one generates every h-multiset, and with it the whole $P_{\nu}^{(q)} \otimes L(\nu)$ (the $L(\nu)$ -factor is irreducible under E, H, f_0). Different q do not interact, so each summand is simple.

For $\nu = 0$, let $p_0 = F^{-1} X u \in P_0$. Then $E p_0 = f_0 p_0 = 0$,

$$h_r p_0 = F^{-1} h_r X u + 2F^{-2} f_r X u = -f_0 e_r p_0, \quad f_j p_0 + \frac{1}{2} f_0^2 e_j p_0 = -\frac{1}{2} h_j X u = -\frac{1}{2} (\lambda_j X + [h_j, X]) u \in M,$$

and $e_r p_0 \in P_2$. Thus every generator maps P_0 into the $\nu \geq 2$ part, so the quotient is the trivial module $P_0 \otimes L(0)$. Since every nonzero submodule of G contains a primitive and hence, by the generation argument, a whole tail $G^{\geq \nu}$, no submodule is simple; the socle vanishes. $\square$

Corollary 9. (i) *In the basic module $M_{-1,1}$ at $\lambda_0 = 0$ with $\lambda_1 \neq 0$: $P_0 = 0$, $\nu_0 = 2$, and $G \supsetneq G^{\geq 4} \supsetneq G^{\geq 6} \supsetneq \dots$. The vector $e_0^2 f_1^2 w_1$ (top layer f_1^2) generates the summand $P_2^{(0)} \otimes L(2)$ of the first quotient.*

(ii) *If $\mu = -2$, $B = L$, and $\lambda_j \neq 0$ for all $1 \leq j \leq L$, then $P_0 = \mathbb{C} w_1$ and the chain starts at $\nu_0 = 0$ with the trivial quotient $\mathbb{C} w_1$.*

5 Scope, smoothness, and the assumptions

The structure of Sections 2–4 uses only the deep region, the integrality $\mu \in \mathbb{Z}$ (otherwise $G = 0$: the pure-fraction recurrence $Ew_m = -m(\mu + m + 1)w_{m+1}$ has no vanishing coefficients and the pole analysis of Lemma ?? forces $\nu \in \mathbb{Z}$), and the PBW machinery. Simplicity of M ($\lambda_L \neq 0$, $c \neq -2$) is not used here; it enters only in the companion addendum, where it guarantees that G_c is the unique maximal submodule of Q_c with simple quotient Q_c/G_c . The two statements are compatible: the head of Q_c is simple while G_c itself has zero socle and infinite length.

In the shallow region $-a \leq c < b$ one has $G_c = 0$, because $E = e_{-c}$ is injective on Q_c . Hence the three questions posed in the addendum—simplicity of G_c , whether G_c is the socle, and its composition length—are answered completely: no, no, and infinite.

Machine checks. The E-action formula (??), the examples of Corollary ??, and the weight-zero identities used in the proof of Theorem ?? were checked by exact symbolic computation (`work/check_gc_sanity.py`, `work/check_gc_filter.py`). These are corroborations, not an independent review or a formal proof.

References

- [1] V. Futorny, X. Guo, Y. Xue and K. Zhao, *Smooth representations of affine Kac–Moody algebras*, arXiv:2404.03855v2 (2024), Theorem 1.1. <https://arxiv.org/abs/2404.03855v2>.
- [2] Project working note, *Boundary and Nonboundary Localization Quotients of Smooth Affine Modules* (September 2026), and its addendum of September 15, 2026: the spectral-flow twist σ_k , the deep-region notation, and the unique-maximal-submodule theorem.